Fraud Detection Pipeline

Summary Report | March 2026

1. Overview

End-to-end fraud detection pipeline: data quality checks, feature engineering, 4 classifiers (Logistic Regression, Random Forest, XGBoost, LightGBM), evaluation with PR-AUC focus, and recall optimisation at 30-60%. Best model: XGBoost (PR-AUC=0.051).

2. Data Quality & Preparation

	Raw	After Dedup	Columns	Fraud Rate
Train	157,306	113,905	48 (44 after label drop)	0.11%
Test	110,998	74,792	44	Unknown

Pipeline: null normalisation (sentinel strings -> NaN) -> deduplication (fraud-priority) -> encoding fixes -> type casting -> identifier removal -> quality report. Temporal hold-out: test starts Apr 2020, train from Feb 2020.

3. Feature Engineering (27 features)

Category	Features	Rationale
Velocity	card_txn_7d (7-day rolling count)	Burst fraud patterns
Card stats	card_avg_amount, amount_deviation	Anomalous amounts
Geographic	card_billing_mismatch, geoip_billing_mismatch	Cross-border fraud
Temporal	hour, dayofweek, is_weekend, day	Time-of-day patterns
Domain	domain_freq (email domain frequency)	Synthetic identities
Amount	log_euramount, merchant_txn_count	Skewness correction
Categorical (8)	cardbrand, cardtype, decline_type, ...	One-hot encoded

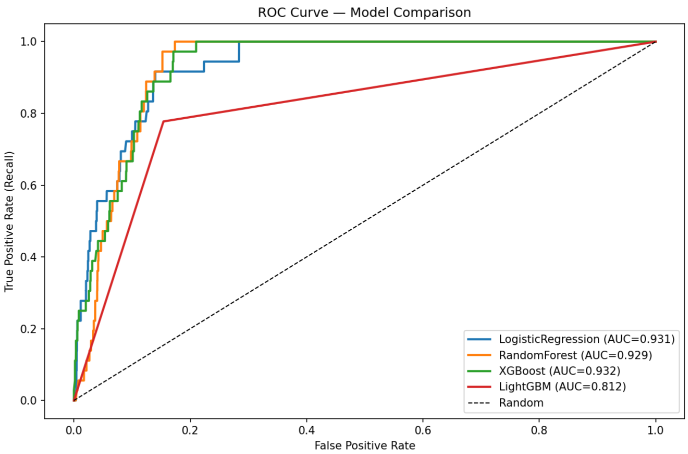
Class imbalance handled via class_weight='balanced' (LR, RF), scale_pos_weight (XGBoost), and is_unbalance=True (LightGBM). Temporal 80/20 split: Train=91,124, Val=22,781 (0.16% fraud).

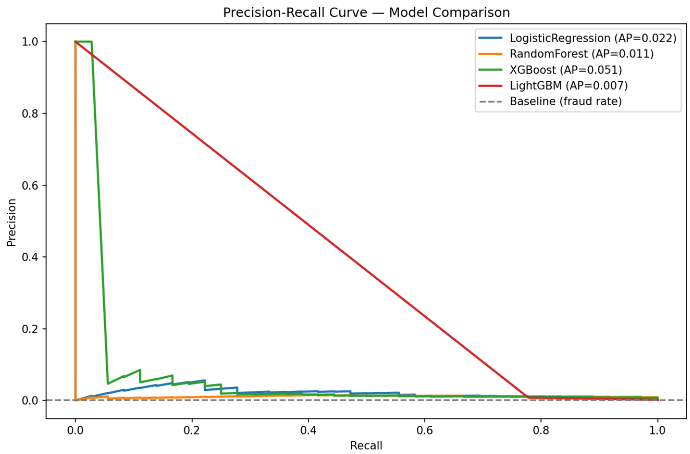
4. Model Results

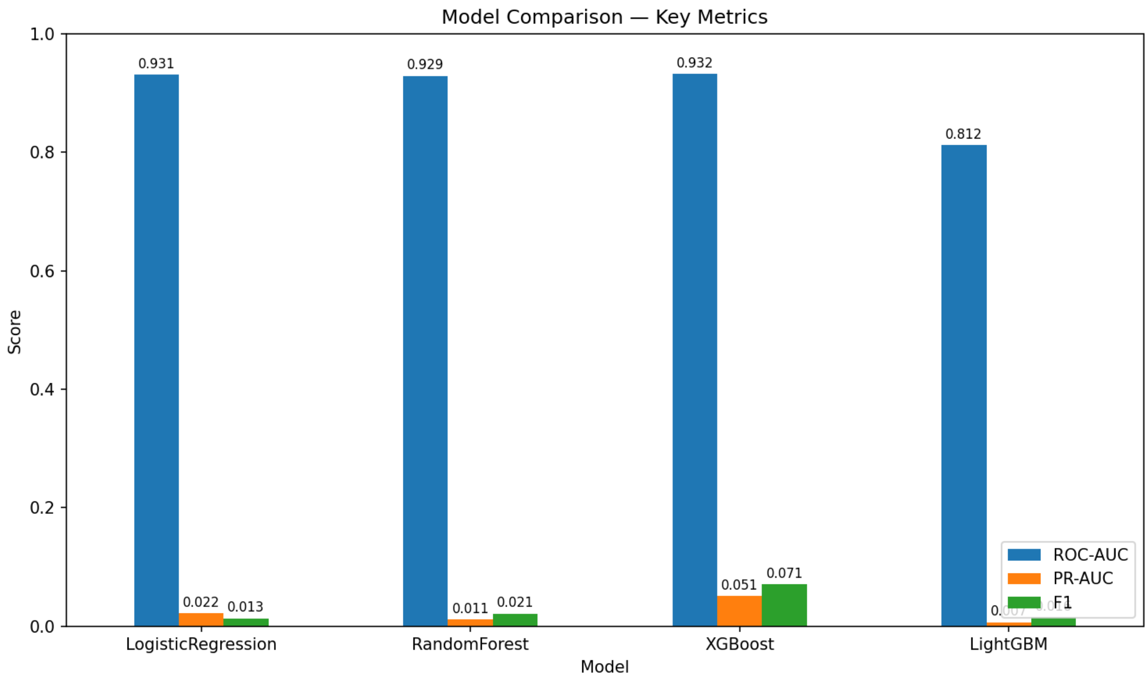
Model	ROC-AUC	PR-AUC	F1	Val Recall	Overfit (PR gap)
LogisticRegression	0.932	0.022	0.013	92%	0.058
RandomForest	0.929	0.012	0.021	75%	0.162
XGBoost*	0.932	0.051	0.071	25%	0.811
LightGBM	0.812	0.007	0.016	78%	0.001

** Best model selected by PR-AUC (most relevant metric for extreme imbalance).*

5. Model Comparison







XGBoost leads on PR-AUC (0.051) despite showing significant overfitting (train PR=0.86 vs val PR=0.05). High ROC-AUC across models (0.81-0.93) masks the difficulty visible in PR-AUC due to extreme class imbalance.

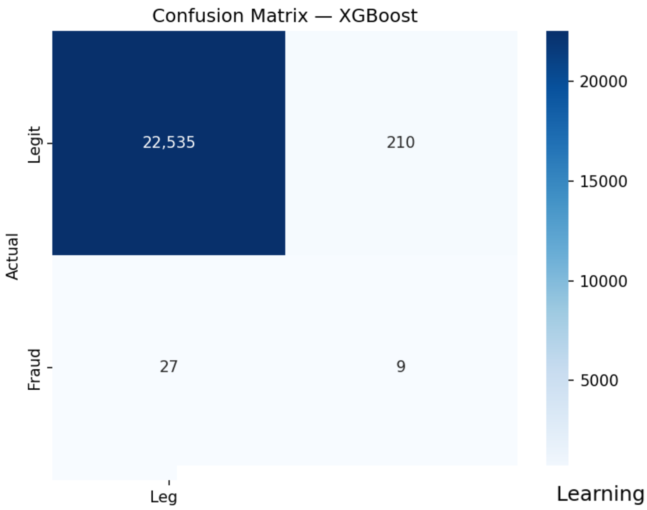
6. Recall Optimisation (30%-60%)

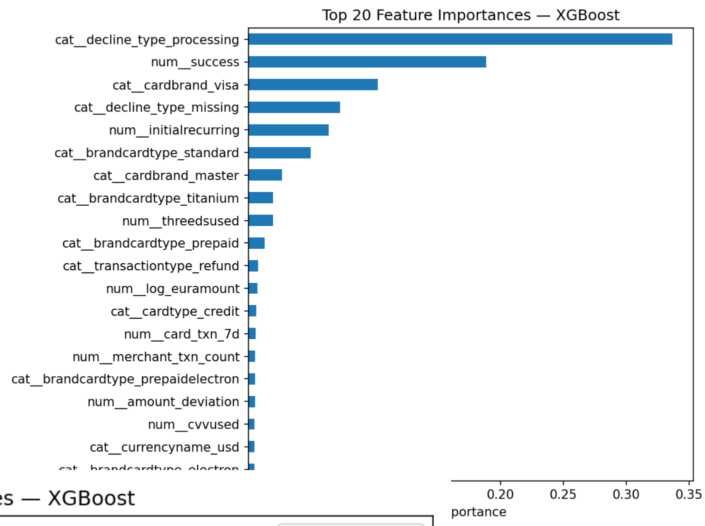
XGBoost (Best Model) -- Recall Operating Points:

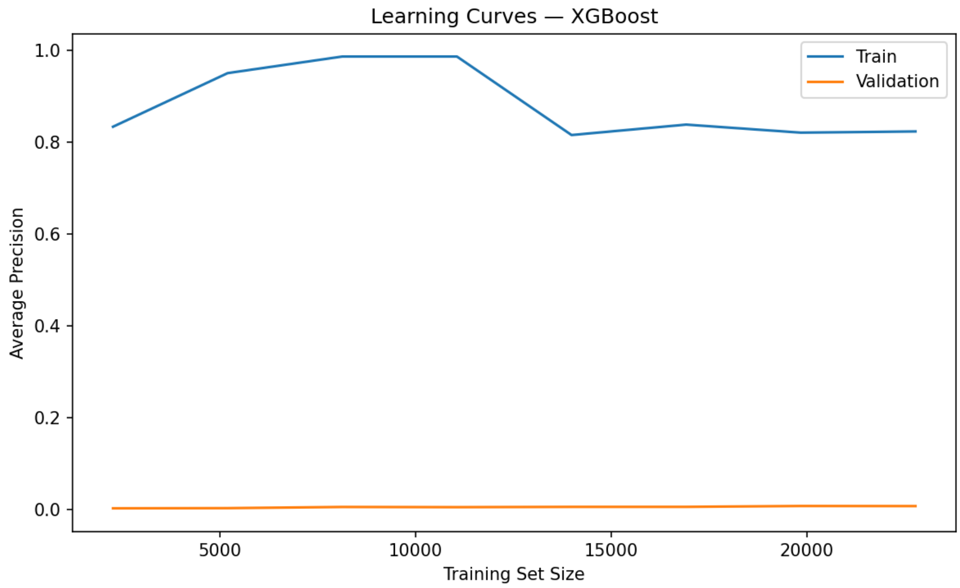
Target	Threshold	Actual Recall	Precision	F1
30%	0.1162	36.1%	0.0195	0.0370
35%	0.1162	36.1%	0.0195	0.0370
40%	0.0742	41.7%	0.0168	0.0324
45%	0.0247	55.6%	0.0140	0.0273
50%	0.0247	55.6%	0.0140	0.0273
55%	0.0247	55.6%	0.0140	0.0273
60%	0.0106	61.1%	0.0116	0.0228

Test predictions: threshold=0.025 (targeting ~50% recall), flagging 3,319/74,792 transactions (4.44%) as potentially fraudulent.

7. Best Model Detail (XGBoost)







8. Conclusions & Next Steps

- XGBoost selected as best model (PR-AUC=0.051, ROC-AUC=0.932)
- Significant